

Lecture 04: Generalization and Model Selection

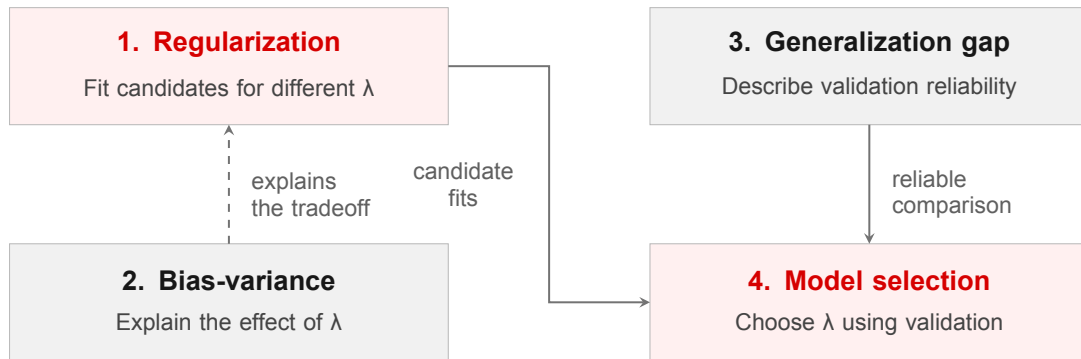
Machine Learning Tutorial

School of Engineering, Westlake University

Fall 2026



Four topics support prediction on new data



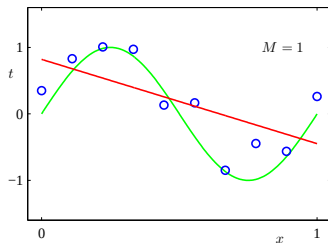
Goal: low true error on new data.

Low training error does not guarantee low true error

Blue points: observed data

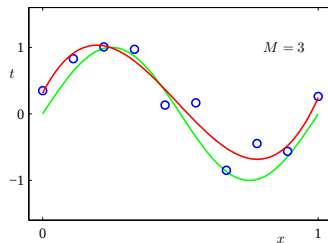
Green: true function

Red: fitted model



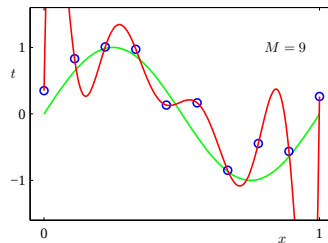
Under-fitting

Misses the main pattern



Closer to the true curve

Follows the overall shape



Over-fitting

Follows training variation

Fitting the observed points closely can still give poor predictions between them.

1.1 Regularization balances training loss and a penalty

Objective

→ Penalty

→ Ridge fit

$$J_{\lambda}(w) = \underbrace{\frac{1}{2N} \|y - Xw\|_2^2}_{\text{training loss}} + \lambda \underbrace{R(w)}_{\text{penalty}}, \quad \lambda \geq 0$$

Term	What it favors
Training loss	Predictions close to observed y
Penalty $R(w)$	Smaller coefficients
Weight λ	The balance between the two

Choose a penalty, then minimize this objective for a given λ .

1.2 Ridge and Lasso use different penalties

Objective

→ **Penalty**

→ Ridge fit

Same training loss. Different preference for the coefficient vector.

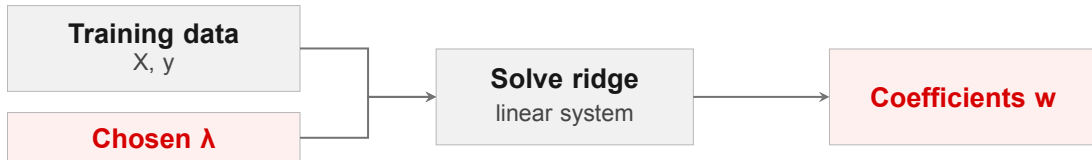
Method	Penalty $R(w)$	Effect on coefficients
Ridge	$\frac{1}{2} \ w\ _2^2$	Shrinks coefficients
Lasso	$\ w\ _1$	Can set some coefficients to zero

Choose the penalty, then fit the coefficients for a given λ .

1.3 A fixed λ determines the ridge fit

Objective

→ Penalty

→ **Ridge fit**

$$J_{\lambda}(w) = \frac{1}{2N} \|y - Xw\|_2^2 + \frac{\lambda}{2} \|w\|_2^2$$

$$\nabla_w J_{\lambda}(w) = 0 \implies (X^{\top}X + N\lambda I)w = X^{\top}y$$

$$w_{\text{ridge}} = (X^{\top}X + N\lambda I)^{-1}X^{\top}y$$

Inverse expression: when the displayed matrix is invertible.

Fitting w for a given λ does not choose the best λ for new data.

2.1 Predictions change across training sets

Repeated fits

→ Error components

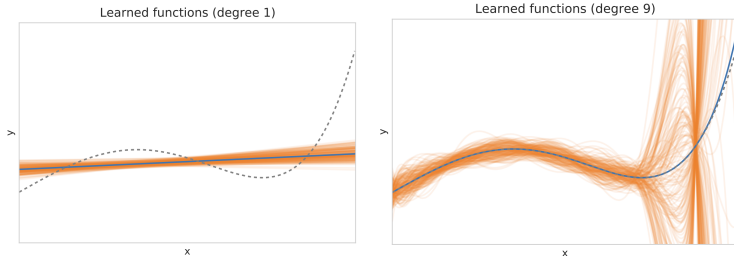
→ Regularization tradeoff

$$y = \underbrace{f(x_0)}_{\text{true response}} + \underbrace{\epsilon}_{\text{noise}} \quad \mathbb{E}[\epsilon] = 0$$

Orange: individual fits

Blue: average fit

Dotted: true function



Noise is independent of the input and training data.

Compare the average fit with the truth, and individual fits with their average.

2.2 Squared prediction error has three components

Repeated fits

→ **Error components**

→ Regularization tradeoff

$$\underbrace{y}_{\text{observed value}} = \underbrace{f(x_0)}_{\text{true value}} + \underbrace{\epsilon}_{\text{noise}} \quad \underbrace{f_{S_{\text{train}}}(x_0)}_{\text{model prediction}}$$

$$\bar{f}(x_0) = \underbrace{\mathbb{E}_{S_{\text{train}}} [f_{S_{\text{train}}}(x_0)]}_{\text{average prediction across training sets}}$$

$$\underbrace{\mathbb{E}_{S_{\text{train}}, \epsilon} [(y - f_{S_{\text{train}}}(x_0))^2]}_{\text{Expected squared prediction error}}$$

$$= \underbrace{\text{Var}(\epsilon)}_{\substack{\text{Noise variance} \\ \text{variation in observed } y}} + \underbrace{(\bar{f}(x_0) - f(x_0))^2}_{\substack{\text{Bias}^2 \\ \text{average prediction vs. truth}}} + \underbrace{\mathbb{E}_{S_{\text{train}}} [(f_{S_{\text{train}}}(x_0) - \bar{f}(x_0))^2]}_{\substack{\text{Variance} \\ \text{prediction vs. its average}}}$$

Fixed x_0 ; zero-mean noise independent of the training set; finite second moments.

The decomposition separates error sources. It does not directly choose λ .

2.3 Regularization can reduce variance and increase bias

Repeated fits

→ Error components

→ **Regularization tradeoff**

As λ increases, the fit places more weight on small coefficients.

Possible change	Error component	Possible effect
Less sensitivity to the sample	Variance	Decreases
Less flexibility to follow f	Squared bias	Increases
Same observation noise	Noise variance	Unchanged

The best λ depends on prediction error. We need a way to estimate it.

3.1 Empirical error estimates true error

Error definitions

→ Independent validation

→ Reliability

f : prediction function ℓ : loss for one example

True error	Empirical error
Average over distribution D	Average over dataset S
$L_D(f) = \mathbb{E}_{(x,y) \sim D}[\ell(y, f(x))]$	$L_S(f) = \frac{1}{ S } \sum_{(x_n, y_n) \in S} \ell(y_n, f(x_n))$
Unknown distribution	Computable from observed data

For fixed f and independent i.i.d. S , empirical error is unbiased.

Which dataset we evaluate on determines what the empirical error tells us.

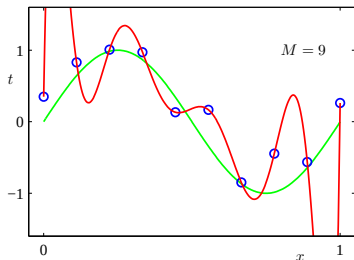
3.2 Independent validation evaluates new-data performance

Error definitions

→ **Independent validation**

→ Reliability

Low training error can mislead

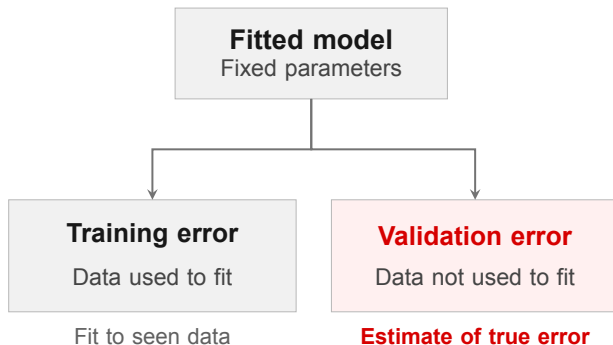


Blue points: training data

Red: fitted model

Green: true function

Evaluate the same fitted model



Validation uses independent data from the same distribution as new data.

3.3 More validation data tightens the error bound

Error definitions

→ Independent validation

→ **Reliability**

Fixed f ; independent i.i.d. validation; loss in $[a, b]$, $a < b$; $0 < \delta < 1$.

$$\Pr \left[\underbrace{|L_{\mathcal{D}}(f) - L_{S_{\text{val}}}(f)|}_{\text{generalization gap}} < \underbrace{\sqrt{\frac{(b-a)^2 \ln(2/\delta)}{2|S_{\text{val}}|}}}_{\text{deviation bound}} \right] \geq \underbrace{1 - \delta}_{\text{confidence level}}$$

With probability at least $1 - \delta$, the gap is below this bound.

Change, holding other terms fixed	Effect on the bound
More validation data	$\text{Bound} \propto \frac{1}{\sqrt{ S_{\text{val}} }}$
Higher confidence (smaller δ)	Larger bound

This controls a probability bound, not the gap of every observed sample.

4.1 Training, validation and test have separate roles

Data roles

→ Candidate comparison

→ Cross-validation

1. Fit candidates

Training set

Fit w for each λ
Learn parameters

Output

Candidate models

2. Select a model

Validation set

Choose λ
Lowest validation error

Output

Chosen λ and model

λ is the hyper-parameter.

3. Evaluate the result

Test set

Evaluate once
Use the chosen model

Output

Final test error

Complete all fitting and model choices before using the test set.

4.2 Validation error selects the hyper-parameter

Data roles

→ **Candidate comparison**

→ Cross-validation

Fit all K candidates on the same training set.

Candidate setting	Fit on training data	Evaluate on validation data
λ_1	f_1	$L_{S_{\text{val}}}(f_1)$
λ_2	f_2	$L_{S_{\text{val}}}(f_2)$
$\vdots$	$\vdots$	$\vdots$
λ_K	f_K	$L_{S_{\text{val}}}(f_K)$

$$\hat{k} = \arg \min_k L_{S_{\text{val}}}(f_k)$$

Compare the candidates using the same validation data.

4.3 Cross-validation averages over different splits

Data roles

→ Candidate comparison

→ **Cross-validation**

Repeat this matrix for each candidate λ .

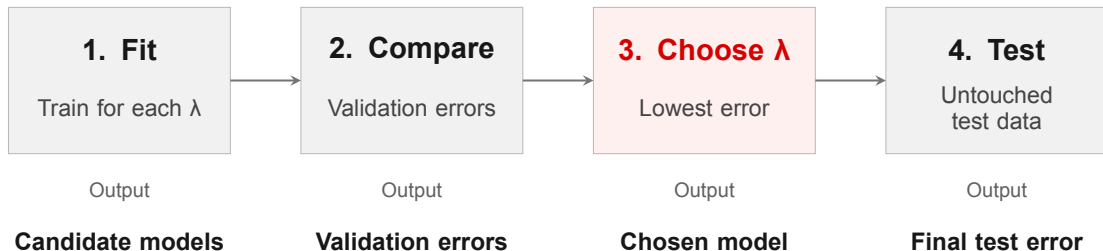
Run	Fold 1	Fold 2	Fold 3	Fold 4
1	Validate	Train	Train	Train
2	Train	Validate	Train	Train
3	Train	Train	Validate	Train
4	Train	Train	Train	Validate

Test set stays outside these folds.

Average the four validation errors, then compare the averages across λ values.

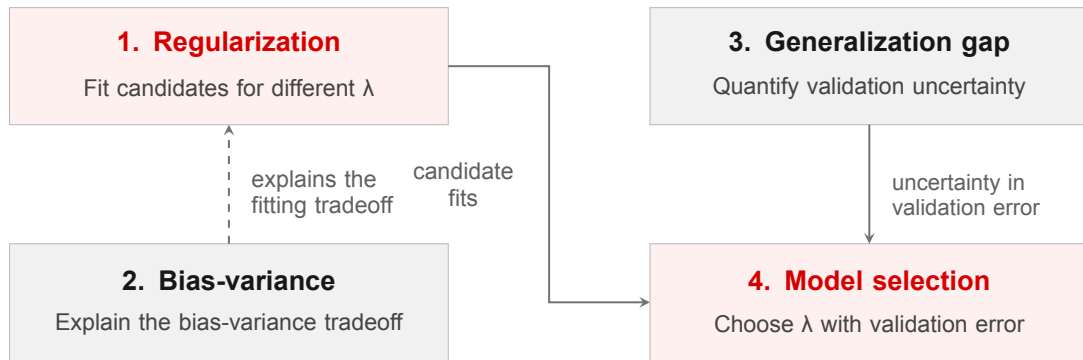
Fit, compare, select and test the final model

Compare all candidates using the same data splits.



Choose λ before evaluating the final model once on test data.

Four topics connect fitting, evaluation and selection



Goal: select a model with low true error on new data.

MAP gives a probabilistic interpretation of ridge

MAP: choose the coefficients with the highest posterior probability.

$$w_{\text{ridge}} = \arg \min_w [-\log p(w \mid X, y)]$$

$$\arg \min_w \left[\underbrace{-\log p(y \mid X, w)}_{\text{fit the observed data}} + \underbrace{-\log p(w)}_{\text{penalize coefficients}} \right]$$

The lecture model gives a squared-error term plus an L2 penalty.